# WasteWise AI — Lean Canvas

*ML-based municipal solid waste generation estimation and collection-priority support platform*

| PROBLEM | SOLUTION | UNIQUE VALUE PROPOSITION | UNFAIR ADVANTAGE | CUSTOMER SEGMENTS |
|---|---|---|---|---|
| • Waste generation varies across cities and years.<br>• Collection planning lacks simple, data-driven estimates.<br>• Existing approaches rely heavily on reports, spreadsheets, and manual judgment. | • Train a Gradient Boosting ML model on official Indian city-level waste data.<br>• Estimate daily waste generation from city, population and year.<br>•Convert estimates into LOW / MEDIUM / HIGH collection-priority support. | •Data-driven waste generation estimates<br>•Simple and explainable ML-based predictions<br>•City-level waste analysis and comparison<br>•LOW / MEDIUM / HIGH collection-priority support Interactive dashboard for decision-making<br>•Uses official Indian government waste data | • Reproducible pipeline built around an official government dataset.<br>• Time-aware evaluation and leakage controls<br>.• Integrated analytics, prediction and explainability in one lightweight dashboard. | •Municipal corporations and urban local bodies<br>•Municipal waste-management departments<br>•Smart-city and urban planning teams<br>•Civic data and analytics teams<br>•Environmental and policy researchers<br>•Academic institutions working on urban sustainability |
| **EXISTING ALTERNATIVES** | **KEY METRICS** | **HIGH-LEVEL CONCEPT** | **CHANNELS** | **EARLY ADOPTERS** |
| • Manual planning and spreadsheet-based estimation• Historical reports and static dashboards<br>• Ad-hoc expert judgment without a consistent ML layer | • Test RMSE: 581.7 t/day• Test $R^2$: 0.920• Test MAE: 405.9 t/day<br>• 119 automated tests passing• 45 cities in final modeling dataset | “A lightweight ML decision-support dashboard for municipal waste planning.” | • Web dashboard / internal municipal tool• Demonstration to civic and planning teams• GitHub / academic project showcase• Internship / institutional deployment | • Data-aware municipal waste teams• Smart-city / urban analytics teams<br>• Researchers working with civic datasets<br>• Student / institutional pilot projects |

| COST STRUCTURE | REVENUE STREAMS |
|---|---|
| • Data preparation and model development• Dashboard hosting / maintenance if deployed• Periodic data refresh and model monitoring• Institutional support and documentation | Prototype / academic use: no direct revenue.Potential future model: institutional licensing, deployment/support contracts, or analytics services for municipalities. |